Introduction to OpenMP

ECM3446/ECMM461: High Performance Computing

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Module Overview

Week 1: Introduction to HPC	Week 5: Floating point arithmetic	Week 8: Task parallelism and manger worker parallelism
Week 2: Introduction to OpenMP	Week 6: Memory and cache	Week 9: Linear algebra and matrices
Week 3: Numerical solutions to PDEs	Week 7: Factors affecting parallel performance	Week 10: Accelerators and GPUs
Week 4: Introduction to MPI		Week 11: Summary

Introduction to OpenMP

- Last week we saw that parallelism is vital to HPC and that HPC applications are usually written in compiled languages
- This week we will look OpenMP which can be used parallelise programs written in C, C++ and Fortran
- SAB Chapter 7: “The Essential OpenMP” covers this week’s material

Syllabus Plan

- Motivation of and introduction to high-performance computing
- Parallel computer architectures: shared-memory and distributed-memory architectures, multi-core processors, Graphics Processing Units (GPUs)
- Parallel programming methods: OpenMP, Message Passing Interface (MPI)
- Floating point arithmetic: floating point model, range, accuracy, exceptions
- Parallel processing algorithm and programming design: domain decomposition, halo exchange, manager-worker, task-based parallelism
- Parallel performance: speed-up, efficiency, parallel overheads and scaling
- Interconnection networks in high performance computers: topologies, latency and bandwidth

Lecture Units

- Unit 2.1: Introduction to OpenMp
- Unit 2.2: OpenMP “Hello World” Demo
- Unit 2.3: OpenMP runtime library
- Unit 2.4: Conditional compilation
- Unit 2.5: Parallelising loops
- Unit 2.6: Variable scoping
- Unit 2.7: Loop carried dependencies

Workshop 2

- Workshop 2 will be on 27th/28th January
- Workshop 2 looks at parallelising loops with OpenMP
- Before the workshop you will need to get familiar with units 2.1-2.7